

The representation of noun–verb distinction in left posterior middle temporal gyrus: evidence from representation similarity analyses

Wenjia Zhang ^{1,2,3,†}, Xuemei Chen ^{1,2,*}, Suiping Wang ^{1,*}

¹Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education, Zhongshan Avenue 55, Guangzhou, Guangdong 510631, China

²School of Psychology, South China Normal University, Zhongshan Avenue 55, Guangzhou, Guangdong 510631, China

³Key Laboratory for Artificial Intelligence and Cognitive Neuroscience of Language, Xi'an International Studies University, South Wenyuan Road 6, Xi'an, Shaanxi 710128, China

*Corresponding authors: Suiping Wang and Xuemei Chen, Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education, 510631, Guangzhou, China. Email: wangsuiping@m.scnu.edu.cn and xuemei.chen@m.scnu.edu.cn

†Wenjia Zhang is the first author (without sharing authorship).

Nouns and verbs are fundamental grammatical building blocks of languages. A key question is whether and where the noun–verb division was represented in the brain. Previous studies mainly used univariate analyses to examine this issue. However, the interpretation of activated brain regions in univariate analyses may be confounded with general cognitive processing and/or confounding variables. We addressed these limitations by using partial representation similarity analysis (RSA) of Chinese nouns and verbs with different levels of imageability. Participants were asked to complete the 1-back grammatical class probe (GCP; an explicit measure) and the 1-back word probe (WP; an implicit measure) tasks while undergoing functional magnetic resonance imaging. RSA results showed that the activation pattern in the left posterior middle temporal gyrus (LpMTG) was significantly correlated with the grammatical class representational dissimilarity matrix in the GCP task after eliminating the potential confounding variables. Moreover, the LpMTG did not overlap with the frontal–parietal regions that were activated by verbs vs. nouns or the task effect (CRP vs. WP) in univariate analyses. These results highlight the role of LpMTG in distinguishing nouns from verbs rather than general cognitive processing.

Key words: grammatical class; noun–verb division; RSA; LpMTG; Chinese.

Introduction

Grammatical class (GC) is one of the most common forms of linguistic information (Robins 1952). Nouns and verbs are fundamental GCs of languages. To test psycholinguistic theories about GC, many neuroimaging studies have examined whether and where the noun–verb division was represented in the brain (Davide et al. 2013). However, no converging evidence was obtained about this issue (Davide et al. 2011). This may be because of the intrinsic characteristics in univariate analyses, which were most used in previous studies. In the current study, by combining the advantage of representation similarity analysis (RSA) and univariate analyses, we tried to tell the whole story about the neural structures of noun–verb distinction in Chinese.

Several psycholinguistic theories have been proposed to explain the representations of nouns and verbs. Broadly speaking, these theories propose that the noun–verb representation is either automatically or nonautomatically accessed in language processing. First, the automatic accounts (e.g. the lexicalist model) assume that the noun–verb distinction is represented as one of the grammatical nodes, which is connected to the lemma node of the word (Pickering and Branigan 1998; Levelt et al. 1999). These lexicalist grammatical nodes are automatically activated when the lemma of a specific word is accessed, given that the spread activation between lemma and syntactic nodes are mostly automatic (Levelt 1989; Roelofs 1992).

Second, the nonautomatic accounts propose that noun–verb distinction is neither lexically specific nor automatically retrieved during word processing. There are two somewhat different versions of the nonautomatic accounts. In the combinatorial model (see Garrett 1984; Ullman et al. 1997), noun–verb distinction is considered as procedural knowledge that is activated when it is needed. In this model, noun–verb representation is independent of other linguistic information, such as semantic relationships, which might be accessed in phrase/sentence level processing. In the emergentist model (see Elman 2004; Farmer et al. 2006), the noun–verb distinction is statistically learned during word processing based on emerging constraints of the semantic context (Elman 2004) or the word's phonological features (Farmer et al. 2006).

To test the above accounts about GC, many previous studies have examined whether and where the noun–verb division was represented in the brain from the neuroscience perspective. Earlier studies mainly used lesion–behavior associations in the patient population to examine whether different brain regions respond to nouns and verbs. These studies suggest that noun processing involves left temporal regions, while verb processing involves left frontal regions (Damasio and Tranel 1993; Chen and Bates 1998; Shapiro et al. 2005). However, recent studies reveal a more complicated picture of the neural representation of noun–verb distinction by using the neuroimaging approach (Davide et al.

2011). The reasons behind these mixed results may come from the following two aspects.

First, many previous studies showed that verb processing elicited more activations in frontal-parietal regions than noun processing (Shapiro et al. 2005; Shapiro et al. 2006; Feng et al. 2020). However, the interpretation of these results is confounded with general cognitive processing because verbs impose greater processing demands than nouns. For instance, retrieving lexical information from long-term memory may be more difficult for verbs than nouns because verbs tend to involve more morphosyntactic processing and variable or context-dependent meanings (Thompson-Schill et al. 1998; Tyler et al. 2004). Therefore, the greater activations for verbs than nouns in frontal-parietal regions may be related to the greater recruitment of attention and working memory resources of verbs rather than the representation of noun-verb division (Yokoyama et al. 2006; Siri et al. 2008).

Second, the interpretation of distinct neural bases between noun and verb processing is also confounded by other variables of the GC contrast (noun vs. verb). Many previous studies tried to overcome this issue by eliminating or manipulating the confounding variables along the noun-verb dimension. However, some studies have focused on different confounding variables such as semantics (Vigliocco et al. 2006; Siri et al. 2008) and imageability (Bedny and Thompson-Schill 2006; Della Rosa et al. 2018). Therefore, it is still unclear about the neural representation of noun-verb distinction, partly because it is difficult to reconcile these results across different studies.

The two issues mentioned above may be accompanied by the intrinsic characteristic in univariate analyses, which were mostly used in previous studies. First, univariate analyses focus on differences in relative activation strength associated with discrete conditions. That is, all words of the same GC are grouped and taken to be the same, and thus the potential fine-grained differences among stimuli were ignored. Therefore, the activation differences in univariate analyses may be more sensitive to core processes that are constant across stimuli such as global engagement in ongoing tasks (Mur et al. 2009). That is, univariate analyses are natively suitable to examine the differences in general cognitive processing between nouns and verbs rather than the content representation of noun-verb distinction. This proposal is consistent with the widely reported results of univariate analyses. That is, verb processing evolved more activations in frontal-parietal regions compared with noun processing because verbs evoked more general cognitive demands than nouns (Yokoyama et al. 2006; Siri et al. 2008).

Second, univariate analyses have intrinsic limitations in controlling confounds along the noun-verb dimension. Specifically, some confounding variables natively co-vary with the noun-verb division to some degree (Moseley and Pulvermüller 2014). For example, nouns are natively more imageable than verbs, especially when the noun and verb are both concrete (Bird et al. 2000; Yu et al. 2011). Therefore, it is difficult (if not impossible) for traditional univariate analyses to construct the ideal noun-verb stimulus with high ecological validity while controlling for potential confounds.

RSA may overcome the above two limitations to some extent. RSA is a type of multivariate data analysis method used in a growing body of neuroimaging studies (Haxby et al. 2014). In the RSA method, the representation dissimilarity matrix (RDM) is the important bridge that connects the content of stimuli with neural activation patterns. A RDM is a square symmetric matrix, indexed by the stimuli horizontally and vertically (in the same

order). Each off-diagonal value indicates the dissimilarity of two different stimuli along a certain dimension of comparison. Two kinds of RDM are normally constructed in RSA. First, theoretical RDMs are created by calculating the dissimilarity between the stimuli, along with some predefined theoretical dimensions of interest. The theoretical RDMs, therefore, quantify the degree to which the two members of a pair of stimuli can be distinguished from each other along a dimension of interest. Second, neural RDMs are created by calculating the multivariate dissimilarity between the neural activation associated with the members of each pair of stimuli. Finally, by calculating the correlation between the target RDM and neural RDMs from multiple brain regions, one can draw inferences about which brain regions best represent the information content of the stimuli such as noun-verb division.

Considering the characteristics of RSA, this method may be more suitable to examine the neural representation of noun-verb division. First, RSA can examine the fine-grained differences among GC and measure the systematic variance distributed across voxels of neural activity (Haxby et al. 2014). In other words, RSA does not directly map the properties of the stimuli to activation strength in the brain. Instead, it builds correspondence between GC similarity among the stimuli and neural similarity among the brain representations of the stimuli. Therefore, RSA is more sensitive to representational differences that are more variably engaged across stimuli such as noun-verb division (Mur et al. 2009; Davis and Poldrack 2013).

Second, partial RSA is natively suitable to isolate the effects of the interested dimension and all other potential confounds (Feng et al. 2017; Zhang et al. 2020). To be specific, experimental words can be set up without any constraints and thus with high ecological validity. These experimental words can be rated along multiple dimensions such as the interested variable of noun-verb division and all potential confounding variables. RDMs are then constructed to isolate the representational content of the experimental words. The target RDM reflects noun-verb distinction, and the multiple control RDMs reflect potential confounding variables. The correlations between the GC RDMs and the neural RDMs are then tested while controlling the RDMs of confounding variables. Therefore, in partial RSA, words with high ecological validity can be used and the impact of all potential confounding variables can be eliminated at the same time.

The current study

In this study, we combined RSA with univariate analyses to comprehensively examine the neural representation of noun-verb distinction. Chinese nouns and verbs with highly ecological validity were constructed. Participants completed one explicit and one implicit task in a within-subject design. The explicit task was the 1-back GC probe (GCP) task, in which participants decided whether the GC of the current word (noun or verb) matched the GC of the preceding word. The implicit task was the 1-back word probe (WP) task, in which participants decided whether the current word was the same as the preceding word.

We first used RSA to examine the neural representation of noun-verb division while eliminating the potential confounds. We predict that the neural representation of noun-verb division would mainly emerge in the GCP task, in which participants explicitly processed the GC information. Second, traditional univariate analyses were used to examine the potential differences between nouns and verbs in each task. This analysis helps reveal the neural basis related to general cognitive processing in the comparison of verb processing with noun processing. By comparing these results with RSA results, we may to some extent

dissociate the neural bases involved in general cognitive processing and representation of noun–verb division (Note we do not adopt an all-or-nothing interpretation between the RSA and univariate analyses. Instead, we highlight the characteristics and take full advantage of each analysis method).

Last, we also examined the task effect of all experimental words. We predict that brain regions involved in general cognitive processing would be activated in the GCP task compared with the WP task because the former task is theoretically more difficult and cognitively demanding for participants. By comparing these results with univariate analysis results of verb vs. noun, we can further validate the activated brain regions related to general cognitive processing.

Methods

Participants

We recruited 20 university students (six males, age range 19–26, mean 20.65) for the functional magnetic resonance imaging (fMRI) experiment. They were all right-handed native Mandarin Chinese speakers with normal or corrected-to-normal vision. No participants reported a history of neurological disorders or reading disabilities. This study was approved by the Psychology Research Ethics Committee of South China Normal University. Written informed consent was obtained before the experiment. All participants were given a small monetary reward at the end of the study to compensate for their time.

Materials

GC and imageability were both considered to construct four types of words: concrete noun, abstract noun, concrete verb, and abstract verb. There were 25 Chinese words for each word type. Each word consisted of two characters. All the verbs were transitive. The 100 experimental words were presented once in each block, and 20 filler words were added in each block because of the task requirement (see “Task and procedure” below).

Thirteen participants were recruited to rate the GC and imageability of experimental words with a 7-point scale. In the GC rating, 1 stands for “extreme noun” and 7 stands for “extreme verb.” In the imageability rating, 1 stands for “extremely low imageability” and 7 stands for “extremely high imageability”. We averaged the scores across participants as the corresponding index for each experimental word. Two-sample *t*-tests showed significant differences between nouns and verbs in GC, $t(98) = 37.53$, $P < 0.001$. Moreover, significant differences were found between concrete words and abstract words in imageability, $t(98) = 13.45$, $P < 0.001$ (see Fig. 1A).

We matched the visual complexity (i.e. the number of strokes in the Chinese characters) word frequency, and emotionality valance between the nouns and verbs. To calculate the visual complexity of each word, we calculated the number of strokes in each of the two characters in each word and then calculated the average number of strokes across the two characters. We also calculated the frequency (logWCount) of each word (Cai and Brysbaert 2010). The emotionality valance index of each word was calculated based on emotionality valance ratings made by 13 participants. They were asked to rate all the experimental words on a 7-point scale, with 1 for “extremely unhappy” and 7 for “extremely happy.” We averaged the scores across participants as the emotionality valance index for each word. Two-sample *t*-tests were conducted between nouns and verbs for the visual complexity, word frequency, and emotionality valance. No significant results were

found, $t_s < 1.07$, $P_s > 0.29$. The information and rating scores for each word type are shown in Table 1.

RDM construction

Four theoretical RDMs were created for the partial RSA. The GC RDM was the target RDM because we were mainly concerned with the neural representation of noun–verb distinction. The visual-pixel, semantic, and imageability RDMs were the control RDMs. Each RDM is a square symmetric matrix, indexed by the 100 experimental words horizontally and vertically (in the same order). Each off-diagonal value indicates the dissimilarity between the specific information associated with two different words. The diagonal values are all zero because they indicate the dissimilarity of information between two identical words.

The GC RDM was created based on the GC rating scores. The distance metric of every two words’ rating scores was calculated using the standard Euclidean distance. These distance values were subsequently assigned to the corresponding elements of the symmetric matrix to create the GC RDM. Therefore, this RDM describes the geometry of the arrangement of the noun–verb dimension information. The imageability RDM was created using the same procedure except that the rating and RDM creation were based on imageability information.

To construct the visual-pixel RDM, we first created 100 figures of all the experimental words. The font, size, and color of the words were the same as they were presented in the experiment. We then loaded them in MATLAB and extracted the pixel matrix of each word. The pixel matrix was further cross-correlated (using Pearson’s correlation) over all 100 words to create a 100×100 correlation matrix. The correlation matrix was finally converted to the visual-pixel RDM ($1-r$).

For the semantic RDM, 13 participants were first recruited to rate the semantic similarity of 100 experimental words. In this rating, we first made pairs of all 100 words to yield 4,950 ($100 \times 100/2 - 50$) word pairs. Participants were asked to rate the semantic similarity of each word pair on a 1–7 scale (1 = extremely dissimilar, 7 = extremely similar). The rating scores were averaged across participants and subtracted from 7. These creating values were subsequently assigned to the corresponding word pairs to create the semantic RDM.

Task and procedure

Each participant completed two tasks in a single functional neuroimaging session. As shown in Fig. 1B, in the 1-back GCP task, participants were asked to press the “1” or “2” button on a response box to indicate whether the GC of the current word matched the preceding word. In the 1-back WP task, participants were asked to respond to indicate whether the current word matched the preceding word. No feedback was provided on either task. The button and order of the tasks were counterbalanced across participants. Every participant completed four blocks (two for each task), each containing 100 experimental words and 20 filler words. Each block was preceded by 10 s and followed by 20 s passive viewing of an asterisk to enhance the estimation of baseline signals.

We used the E-Prime 2.0 software package (Psychology Software Tools, Pittsburgh, PA) to present the stimuli and collect behavioral data. The trials in each block were presented in a rapid event-related design in a pseudorandom sequence. At the beginning of each trial, a visual fixation cross was displayed on the screen (0.5 s), followed by a blank interval (0.3 s). Then a word was presented (1.2 s), and participants were asked to respond according to the task requirements. The Chinese characters were presented in Kai font in black against a light gray background. Last,

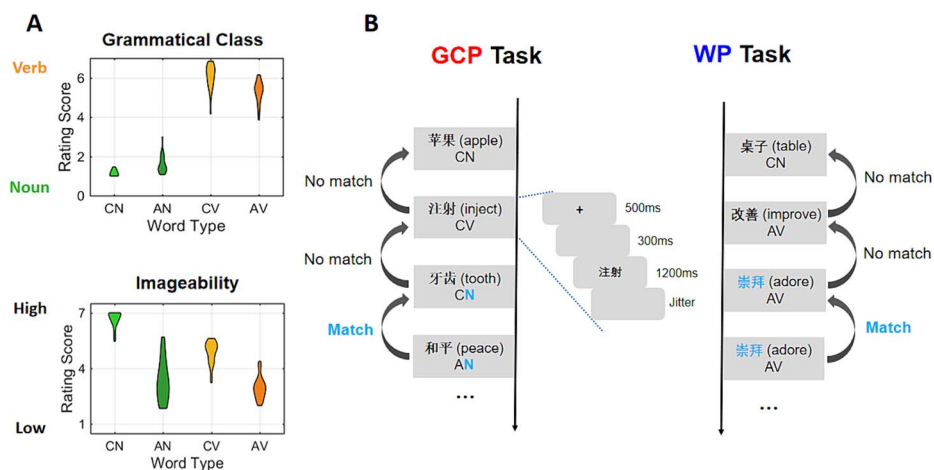


Fig. 1. The information of experimental words and procedure. A) Distribution (violin plot) of GC and imageability rating scores of four types of words. B) Illustration of the task and procedure. In the 1-back GCP task, the GC of the words “牙齿(tooth)” and “和平(peace)” were both nouns. In the 1-back WP task, the word “崇拜(adore)” was matched. The middle graph shows the procedure for each trial. Note that the word-type labels and English translations are for illustration. Only the Chinese word was presented in the fMRI experiment. GCP = grammatical class probe; WP = word probe. CN = concrete noun; AN = abstract noun; CV = concrete verb; AV = abstract verb.

Table 1. The information and rating scores for each word type.

Word type	Example	GC (Noun–Verb)	Imageability	Strokes	Frequency	Emotion Valence
CN	苹果 (apple)	1.06 (0.11)	6.67 (0.43)	7.68 (1.32)	2.20 (0.67)	4.35 (1.21)
AN	和平 (peace)	1.59 (0.46)	3.31 (1.10)	8.22 (1.58)	2.33 (0.61)	4.30 (1.31)
CV	注射 (inject)	6.07 (0.62)	4.86 (0.61)	8.04 (2.35)	2.16 (0.45)	3.92 (0.94)
AV	改善 (improve)	5.30 (0.57)	2.95 (0.61)	7.90 (1.81)	2.30 (0.42)	4.21 (1.36)

a blank interval (average: 3 s) was presented as a jitter between every two trials to better estimate the hemodynamic response. Therefore, the average trial duration was 5 s. Each participant performed 20 practice trials to become familiarized with the procedure before the fMRI scanning.

MRI data acquisition and preprocessing

All imaging data were acquired on a 3.0 T Siemens Trio Tim scanner. We adopted the coronal slice orientation scanning for the functional imaging runs to minimize signal loss and distortion in temporal lobe regions due to the magnetic susceptibility artifact (Axelrod and Yovel 2013). Specifically, the functional images were acquired with a T2*-weighted gradient echo-planar imaging (EPI) pulse sequence (TR = 2000 ms, TE = 30 ms, flip angle = 90°, 37 slices, FOV = 224 mm × 224 mm, in-plane resolution = 3.5 × 3.5, slice thickness = 3.5 mm with 0.7 mm gap). T1-weighted high-resolution structural images were acquired between two tasks for each participant. Specifically, these images were acquired with a magnetization-prepared rapid acquisition gradient echo (MP-RAGE) sequence and sagittal slice orientation (176 slices, TR = 1900 ms, TE = 2.53 ms, FOV = 256 mm × 256 mm, flip angle = 9°, voxel size = 1 × 1 × 1 mm³, duration = 4 min 26 s).

We used SPM12 (Wellcome Department of Imaging Neuroscience, London, UK; www.fil.ion.ucl.ac.uk/spm/) for MRI data preprocessing and univariate analyses. All functional images were corrected for head motion and realigned to the first functional image for each participant. The mean functional images were

co-registered with the structural image and then segmented for each participant. The realigned functional images were smoothed with a Gaussian kernel of 6 mm full width at half maximum. The resulting smoothed native-space functional images were used for the univariate analyses. The unsmoothed native-space functional images were used for the RSA.

RSA

We used the CoSMo RSA toolbox (Oosterhof et al. 2016) and customized MATLAB functions for this analysis. We first conducted the whole-brain searchlight RSA for target GC RDM in each task. Firstly, experimental word-based statistical brain activation maps of each task were generated with the general linear model (GLM) approach for each subject. The interest regressors of the GLM were each experimental word in each task and the regressors of no-interest were filler words and six motion realignment parameters. High-pass filtering was used with a time constant of 128 s to reduce the influence of low-frequency noise. The 100 t-statistic maps of experimental words were further generated in each task. We then assigned each voxel with its nearest neighbor voxel to construct a spherical searchlight (~100 voxels). The t-statistic values of each spherical searchlight were normalized and further extracted to calculate distances between each pair of words (using 1-Pearson correlation). This created the RDMs based on neural activation patterns (see Fig. 2).

The neural RDMs were then correlated with the target GC RDM. The resulting similarity values were mapped onto the central

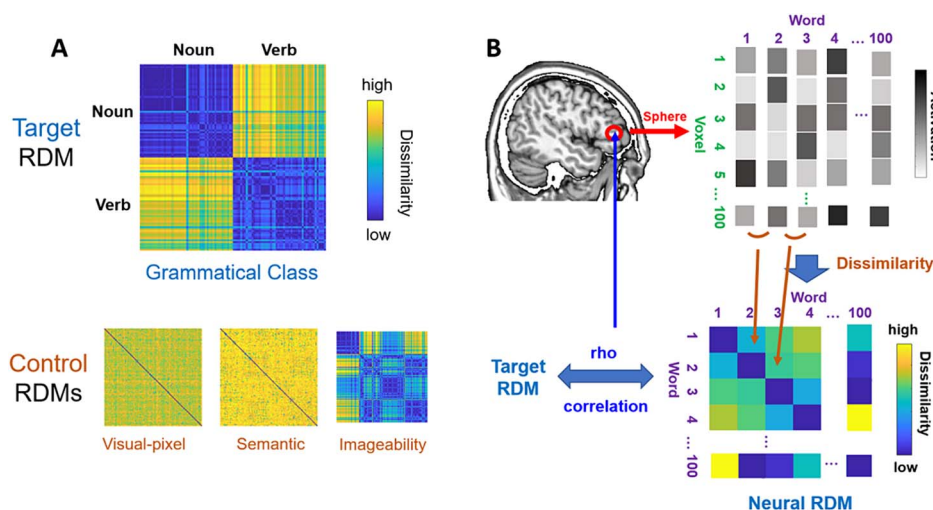


Fig. 2. The four theoretical RDMs and illustration of searchlight RSA. A) The target RDM is the grammatical class RDM. The visual-pixel, semantic and imageability RDMs are the control RDMs. B) In each sphere searchlight, the t-statistic values of 100 voxels for each experimental word were first extracted. These values were used to calculate the dissimilarity between the members of each pair of words, which created the neural pattern RDM. The neural pattern RDM was then correlated with the target RDM. The resulting rho value was finally mapped back to the central voxel of each sphere searchlight.

voxel of each sphere searchlight. The center of the searchlight volume was moved one voxel at a time within the gray matter of the brain. The similarity values of all voxels were r-to-z transformed using the *atanh'* algorithm. The similarity map of each participant was normalized to the Montreal Neurological Institute (MNI) template space. We combined the maps across participants and used one-sample t-tests to examine significant brain regions in each task. The analysis results were corrected for multiple comparisons using voxels significant at 0.005, with an FWE correction at the cluster level of 0.05. The BrainNet View toolbox was used to show the significant activations as t-value maps onto the MNI surface brain template (Xia et al. 2013).

As shown in Fig. 3A, in both the GCP and WP tasks, the GC RDM was significantly correlated with the activation pattern in the left posterior middle temporal gyrus (LpMTG), although the specific significant clusters were a bit different. We then extract the intersection and union voxels of these two significant clusters to define the intersection and union regions of interest (ROIs), respectively. These two ROIs were further used for the ROI-based (partial) RSA in each task. In the ROI-based (partial) RSA, GC RDM was still the target RDM, whereas we used no control RDM (NC) or all three control RDMs (visual-pixel, semantic, and imageability, C).

The analysis protocol of the ROI-based (partial) RSA is similar to that in the searchlight RSA. The main difference was that we used the voxels within each ROI to create the neural RDM and directly computed the correlation coefficient between this RDM with the GC RDM. In the partial RSA, all the RDMs were first vectorized. The partial correlation coefficients were calculated between the GC RDM and neural RDM while controlling all three control RDMs. This partial RSA approach can reveal the neural representations of GC that are linearly independent of other confounding variabilities.

Univariate analyses

GLM was used for the individual-level univariate analyses. In each block, the model included two interest regressors for nouns and verbs, seven regressors of no-interest for filler words, and six motion realignment parameters. High-pass filtering was used with a time constant of 128 s as in the RSA. Voxel-wise parameter

estimates (beta values) of regressors were estimated after fitting the GLM. The parameter estimates were then used to generate statistical (contrast) maps. The contrast maps of the GC effect (noun vs. verb and verb vs. noun) were first generated in each task. Moreover, the contrast maps of the task effect (GCP vs. WP and WP vs. GCP) of all experiment words were also generated.

For the group-level analyses, the contrast images generated from the individual level were first normalized onto the MNI template space using the parameters obtained in the segmentation step. The normalized contrast maps were then combined across participants and one-sample t-tests were used to examine significant brain regions. We used the same correction threshold for multiple comparisons as that in the RSA (voxel: $P < 0.005$, cluster: FWE, $P = 0.05$). We first examined the potential differences between nouns and verbs in each task. Moreover, we also examined the task effect of all experimental words.

Results Behavior

As in the univariate analyses, we first examined the difference between nouns and verbs in each task. The paired sample t-tests showed that verbs had significantly longer reaction time (RT) than nouns in the GCP task, $t(19) = 5.19$, $P < 0.001$ (see Fig. 4A). No significant difference was found in the WP task, $t(19) < 1$. Moreover, the results of task effect showed that the participants showed significantly lower accuracy (ACC), $t(19) = 2.71$, $P < 0.05$, and longer RT, $t(19) = 10.89$, $P < 0.001$ in the GCP task compared with WP task (see Fig. 5A).

RSA

As shown in Fig. 3A, the searchlight RSA results showed that the GC RDM was significantly correlated with the activation pattern in the LpMTG in both the GCP and WP tasks. The significant cluster in the GCP task contained 84 voxels (peak intensity: $t = 5.02$; peak MNI coordinates: $x, y, z = -51, -58, 1$). The significant cluster in the WP task contained 96 voxels (peak intensity: $t = 4.03$; peak MNI coordinates: $x, y, z = -54, -55, 5$).

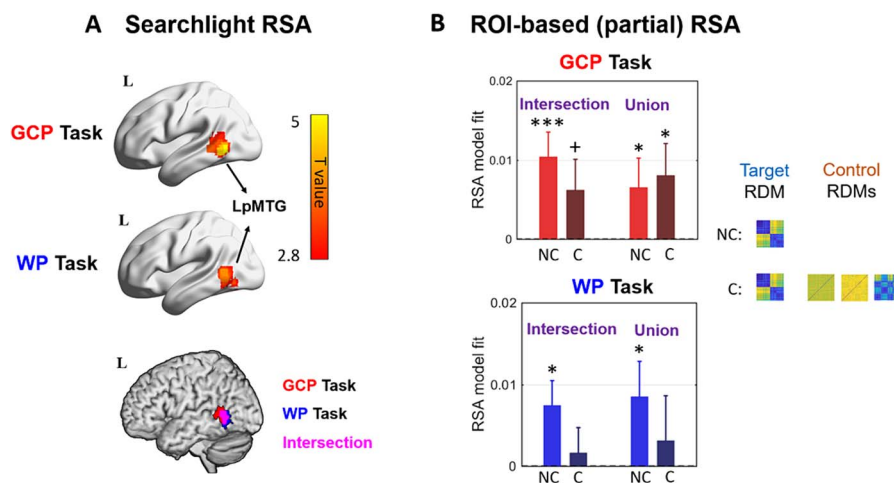


Fig. 3. Searchlight and ROI-based (partial) RSA results. A) The searchlight RSA results of the GC target RDM in each task. The illustrations at the top and middle show significant brain regions (in each task) projected onto the surface brain template. The illustration at the bottom shows significant (binary) brain regions (both tasks) projected onto the brain template. B) The ROI-based (partial) RSA results in each task. NC, RSA with no control RDM; C) partial RSA with all three control RDMs (visual-pixel, semantic, and imageability). GCP = grammatical class probe; WP = word probe. L = left hemisphere; LpMTG = left posterior middle temporal gyrus. *** $P < 0.005$; * $P < 0.05$; + $P < 0.1$.

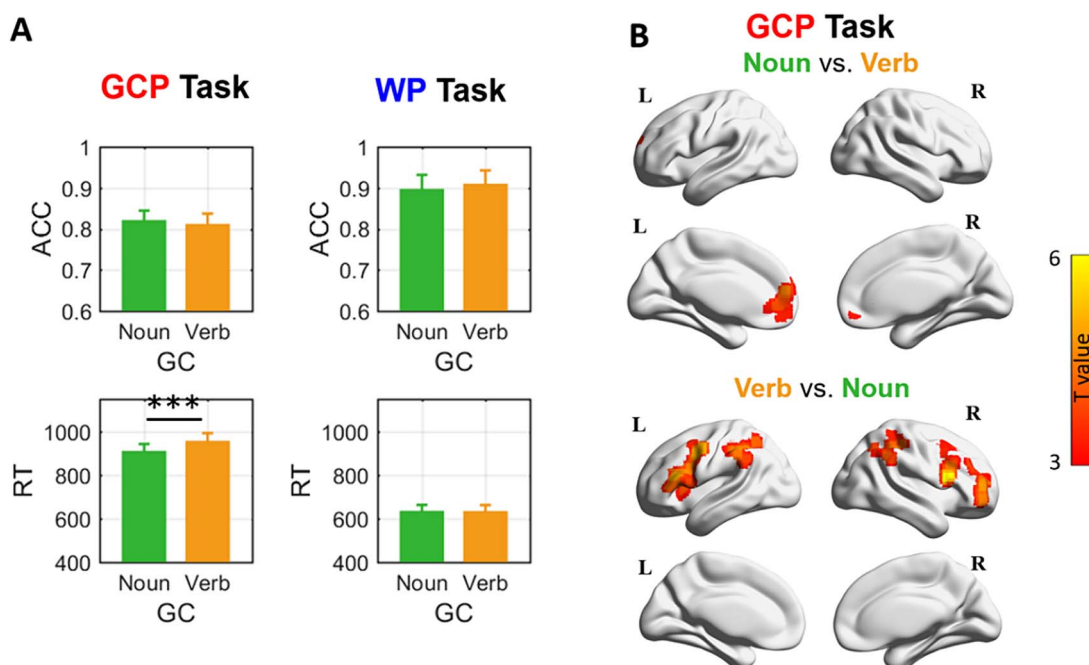


Fig. 4. The behavioral and univariate analysis results of noun-verb distinction in each task. A) The ACC and RT results. B) The searchlight univariate analyses result in the GCP task. No significant result was found in the WP task. GCP = grammatical class probe; WP = word probe. ACC = accuracy; RT = reaction time. L = left hemisphere; R = right hemisphere. *** $P < 0.005$.

As shown in Fig. 3B, the ROI-based (partial) RSA was used in both the intersection and union LpMTG ROIs in each task. In the GCP task, the GC RDM was significantly correlated with the neural pattern in the intersection ROI, $t(19) = 3.35$, $P < 0.005$. This effect was marginally significant with all three control RDMs, $t(19) = 1.61$, $P < 0.1$. For the union ROI, the GC RDM was significantly correlated with the neural pattern without any control RDMs, $t(19) = 1.77$, $P < 0.05$. This effect was still significant with all three control RDMs, $t(19) = 2.01$, $P < 0.05$. In the WP task, the GC RDM was significantly correlated with the neural pattern in both the intersection and union ROIs, $t_s > 1.98$, $P_s < 0.05$. However, the partial RSA showed no significant effect in any ROI with three control RDMs in the WP task, $t_s < 1$.

Univariate analyses

The univariate analyses of the GC effect in each task showed significantly activated clusters in the GCP task. To be specific, the bilateral ventromedial prefrontal cortex (vmPFC) was more activated in the contrast of noun vs. verb, and bilateral frontal-parietal areas were more activated in the contrast of verb vs. noun (see Fig. 4B and Table S1). No significant result was found in the WP task. To fully compare the RSA and univariate analyses, we presented the significant clusters of noun-verb distinction between these two analyses on the same brain template. As shown in Fig. S1, the significant cluster of LpMTG in RSA did not overlap with the activated regions in noun vs. verb or verb vs. noun contrasts in univariate analyses.

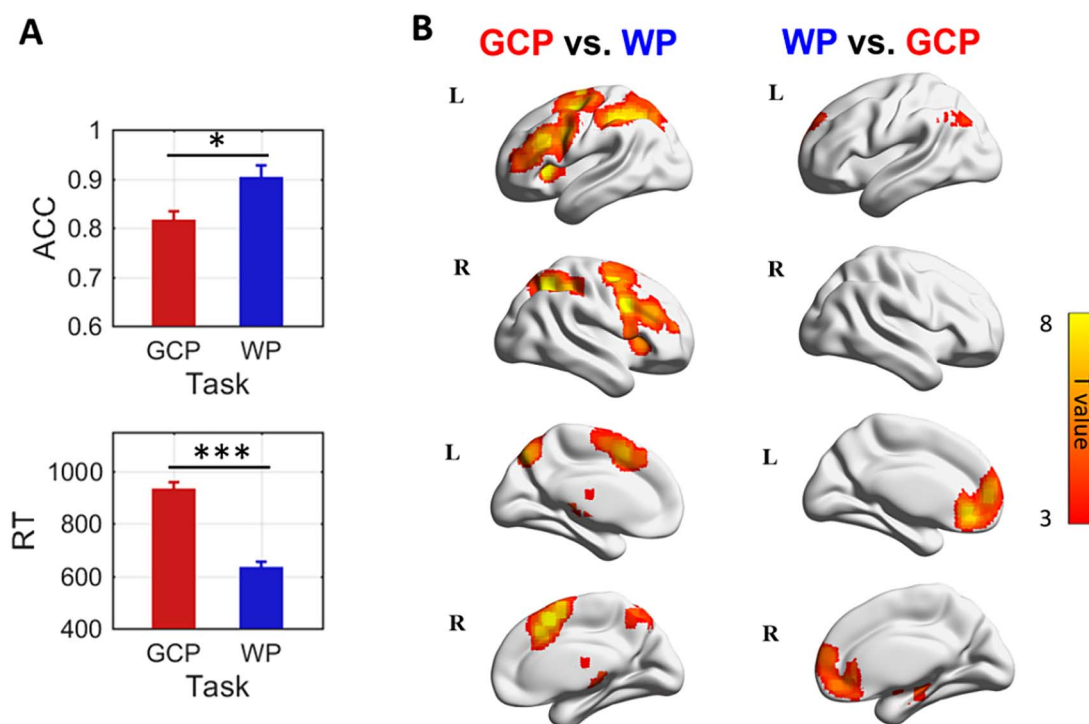


Fig. 5. A) The ACC and RT results of the task effect. B) The searchlight univariate analyses result of task effect. GCP = grammatical class probe; WP = word probe. ACC = accuracy; RT = reaction time. L = left hemisphere; R = right hemisphere. *** $P < 0.005$; * $P < 0.05$.

Moreover, the searchlight univariate analyses of the task effect showed greater activations in the bilateral frontal-parietal areas in the GCP task vs. WP task and greater activations in bilateral vmPFC and LIPL in the WP task vs. GCP task (see Fig. 5B and Table S2). To compare the results of GC and task effects in univariate analyses, we presented the significant clusters of GC and task effects on the same brain template. As shown in Fig. S2, the bilateral frontal-parietal areas largely overlapped between the GCP task vs. WP task and verbs vs. nouns in the GCP task. Moreover, the bilateral vmPFC largely overlapped between the WP task vs. GCP task and nouns vs. verbs in the GCP task.

Discussion

The current study combined RSA with traditional univariate analyses to examine the potential neural representation of noun-verb distinction. The partial RSA results showed that the GC RDM was significantly correlated with the activation pattern in the LpMTG in the GCP task when controlling for potential confounds. The results suggest that LpMTG is involved in the neural representation of noun-verb distinction. Moreover, the univariate analyses showed that verb processing elicited more activation than noun processing in the frontal-parietal regions, which largely overlapped with the activated regions in the GCP task than the WP task. These results suggest that general cognitive processing is mainly involved in frontal-parietal regions. Taken together, our results suggest that the neural representation of noun-verb distinction is involved in LpMTG, which is unrelated to the general cognitive processing or potential confounds. We further discuss the implications of these findings below.

Implications for the psycholinguistic accounts of GC

The multiple models that have been proposed to explain the potential noun-verb distinction in the brain can be broadly

grouped into two accounts: automatic vs. nonautomatic. Our partial RSA results showed that the neural separability of nouns and verbs was absent in the implicit WP task after controlling the confounding variables. This finding is inconsistent with the automatic account of the lexicalist model which proposes that the representation of noun-verb distinction is automatically retrieved and should be detectable on both implicit and explicit tasks (Pickering and Branigan 1998; Levelt et al. 1999). Instead, the RSA results were consistent with the nonautomatic account, which proposes that noun-verb distinction is not automatically retrieved and will only be detectable in explicit tasks.

Although the results were generally more consistent with the nonautomatic account than the automatic account, there was an inconsistency between the two versions of the nonautomatic account. Specifically, our finding of noun-verb distinction might be better explained with the combinatorial model than the emergentist model. The combinatorial model predicts that independent representations of GC will be accessed when it is necessary. In the case of the current study, noun-verb distinction should be accessed in the explicit when it is needed, but not in the implicit task. Our findings supported this explanation. The emergentist model goes further in predicting that the representations of GC on explicit tasks will disappear when the confounding variables (e.g. semantics) are controlled. Our results did not support this explanation.

However, it is worth noting that the GCP task is an explicit task, in which participants were asked to actively distinguish nouns and verbs. Moreover, we did not directly compare combinatorial and noncombinatorial processing of words. This type of comparison would provide direct evidence about the test of combinatorial vs. emergentist explanations. Therefore, the result based on the explicit GCP task should be interpreted with caution. Further studies with more tasks and levels of stimuli (such as phrases and sentences) are needed to examine the different versions of the nonautomatic account of noun-verb distinction.

The distinct roles of LpMTG and frontal-parietal regions

Our study revealed that the neural representation of noun-verb division is involved in LpMTG. This is consistent with previous studies, which showed that LpMTG responded differently to nouns and verbs by using univariate analyses (Marina et al. 2008; Bedny et al. 2014; Mireia et al. 2014). However, our study takes a step forward about the more specific role of LpMTG by taking advantage of RSA. First, beyond the potential general cognitive processing, the LpMTG indeed represents the noun-verb division because the RSA is more sensitive to the content representation among stimuli than the univariate analyses. Second, the partial RSA results provided clear evidence that the role of LpMTG in noun-verb division is unrelated to the potential confounding variables.

Unlike the RSA results, the results of univariate analyses showed greater activation in bilateral frontal-parietal regions in the contrast of verbs vs. nouns in the GCP task. This finding replicated the results of other studies which found greater activations for verbs than nouns in frontal-parietal regions (Palti et al. 2007; Feng et al. 2020). Meanwhile, the RT was longer for processing verbs than nouns in the GCP task. This further suggests that verb processing was more cognitively demanding than noun processing in the GCP task. Therefore, the frontal-parietal regions, at least in part, reflect general cognitive processing, such as working memory and attention allocation, which supports the GCP task (Petersen and Posner 2012). This argument is also supported by the nonsignificant results in the WP task. That is, the processing of nouns and verbs involves similar levels of cognitive demands, which induces nonsignificant results in the WP task.

The task effect in our study is especially helpful in identifying the brain regions that are associated with task-dependent strategies controlled by cognitive demands. This is because identical experimental words were used in the two tasks, strictly controlling for lexical differences. The results of the task effect also showed activations in bilateral frontal-parietal regions during the GCP task relative to the WP task. Meanwhile, the behavioral results showed lower ACC and longer RT in the GCP task relative to the WP task. These results support our proposal that the GCP task required more cognitively demanding for the participants (This is also supported by the activated default mode network (such as vmPFC) during the WP task relative to the GCP task (see Fig. 5B), which suggests that WP task is less cognitively demanding than the GCP task). Therefore, the results of the task effect provide further evidence that the activations in bilateral frontal-parietal regions, at least partially, reflect general cognitive processing.

It is worth noting that in addition to being involved in general cognitive processing, frontal-parietal regions may also reflect language-specific processing. For example, verbs refer more to actions whereas nouns refer more to objects. Meanwhile, participants were asked to explicitly engage with the noun-verb division in the GCP task. Therefore, the frontal-parietal regions may also be related to semantic differences (such as actions vs. objects) associated with nouns and verbs (Xia and Peng 2022). However, note the goal of our study was not to dissociate the roles of the frontal-parietal regions in general cognitive processing or language-specific GC processing. Instead, by demonstrating that general cognitive processing mainly involved frontal-parietal regions (not LpMTG), we propose that the role of LpMTG in the representation of noun-verb division is language-specific and unrelated to general cognitive processing.

Additional considerations and future research

There are four considerations in terms of stimuli, experimental design and analyses in the current study. First, the dimension along the noun-verb division of words is natively continuous (continuously from full nouns to full verbs) rather than dichotomous. This is more salient in our study, which used Chinese words as stimuli (Li et al. 2004; Zhao et al. 2017). To be specific, as in many languages, verbs are sometimes GC-ambiguous, which can be used as a verb or a noun in different sentences. For instance, the verb “改善 (gǎi-shàn, improve)” can be used as a noun in the sentence “环境的改善很重要 (The **improvement** of the environment is important)”. In this case, the GCP task might require more cognitive demands than the WP task, leading to longer RT, lower ACC, and greater activation in bilateral frontal-parietal regions. However, it is worth noting that the significant differences between nouns and verbs in the GC rating support the noun-verb division of our stimuli (see the *Materials* section and Fig. 2A). Our goal is to examine the neural bases of noun-verb division using words with high ecological validity. The fine-grained differences among GC (noun or verb) exactly highlight the advantages of our RSA method.

Second, as two fundamental GCs, nouns and verbs differ systematically at various linguistic levels (semantic completeness, morphology, etc.). For instance, verbs have a more complex argument structure than nouns in our study. In other words, the meaning of nouns (such as apple) is already complete, whereas the meaning of verbs (such as inject) is less complete and requires a direct object. Moreover, verbs are in general more complex than nouns in morphology. Any of these differences may contribute to the GC effects drawn from neuropsychological evidence. Future studies can further dissociate these linguistic differences between nouns and verbs and examine corresponding neural bases.

Third, the within-participants design in our study helps avoid potential individual differences. However, in this case, the explicit GCP task may interfere with the implicit WP task for half of the participants because the order of the tasks was counterbalanced across participants. To be specific, participants may also pay attention to GC in the implicit WP task because of the influence of the preceding GCP task. In order to investigate this possibility, we split the data into two datasets according to the task order (i.e. the implicit WP task in first or second order) and then conducted RSA of the implicit task for both datasets. The results showed no significant clusters in both the datasets (see *Supplementary Materials*). Additionally, there was no significant effect of noun-verb distinction with the implicit task in the univariate analyses of our study. Altogether, these results suggested that the explicit task does not significantly influence the implicit task in our study. Further studies can adopt a between-participants design or fixed task order (implicit task before explicit task) to completely avoid this potential limitation.

Last, although RSA can be applied to conventional experimental designs, the benefits of RSA will be greatest for experimental designs targeting item-level differences with high-resolution measurements (Kriegeskorte et al. 2008). Note the experimental words in our study can be roughly grouped into two categories (nouns and verbs). However, the dimension along the noun-verb division of words is natively continuous rather than dichotomous. Therefore, the advantages of RSA are fully taken because of the fine-grained differences among GC. Altogether, RSA is more suitable for item-level experimental design and examining the neural mechanism related to continuous variables.

Conclusion

A key question about GC is whether and where the noun–verb division was represented in the brain. The current study revealed that LpMTG involves the neural representation of noun–verb division in the brain. Moreover, combining the advantage of RSA and univariate analyses, our study further suggests that this role of LpMTG is unrelated to the general cognitive processing or potential confounding variables.

Acknowledgements

We have no known conflict of interest to disclose.

Author contributions

Wenjia Zhang (Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Xuemei Chen (Conceptualization, Funding acquisition, Investigation, Validation, Visualization, Writing—original draft, Writing—review & editing), and Suiping Wang (Conceptualization, Funding acquisition, Supervision, Validation, Writing—review & editing).

Supplementary material

Supplementary material is available at *Cerebral Cortex* online

Funding

This work was supported by the National Natural Science Foundation of China (3217051 to Suiping Wang), Key Research and Development Program of Guangdong, China (2023B0303010004 to Suiping Wang), Guangdong Basic and Applied Basic Research Foundation (2022A1515110419 to Xuemei Chen), Social Science Foundation in Shaanxi Province of China (2022K015 to Wenjia Zhang), and Natural Science Foundation in Shaanxi Province of China (2023-JC-QN-0232 to Wenjia Zhang).

Conflict of interest statement: None declared.

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